

Huan Wang

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Education

Ph.D.	Yale University , Computer Science	2008–2013
	- Advisor: Prof. Daniel Spielman, Prof. John Wright - Research on clustering, regression, and dictionary learning	
M.Phil.	The Chinese University of Hong Kong , Information Engineering	2005–2007
	- Advisor: Prof. Xiaoou Tang, Prof. Shuicheng Yan, Prof. Jianzhuang Liu - Research on face recognition, manifold learning, subspace learning, semi-supervised learning	
B.Eng.	Zhejiang University , Information Engineering, Chu Kochen Honors College	2000–2004

Professional Experience

Senior Director	Salesforce Research	Palo Alto, CA Feb 2025 – Present
	Large Action Models (xLAM), AI Agents, Reasoning, Planning, SWE Agents, Agent-force	
Director	Salesforce Research	Palo Alto, CA Nov 2021 – Feb 2025
	Led teams in AI for Operational Research (AIOps), AI for Software, Conversational AI, Time-Series Anomaly Detection, Uncertainty Estimation, and Data Hardness Evaluation	
Senior Manager	Salesforce Research	Palo Alto, CA Nov 2019 – Nov 2021
	Deep learning theory and applications, reinforcement learning, multi-task learning, language modeling, multilingual NER, knowledge graph	
Senior Research Scientist	Salesforce Research	Palo Alto, CA 2018 – 2019
	Deep learning research in large-scale language modeling and vision-language integration	
Senior Applied Researcher	Microsoft AI Research	Sunnyvale, CA 2015 – 2018
	- Developed deep learning systems for recommendation, ranking, and intelligent Q&A - Improved web and local search relevance using neural embeddings	
Research Scientist	Yahoo! Labs	New York, NY 2013 – 2015
	- Designed large-scale ML algorithms for search ads prediction and account security prediction - Leveraged Hadoop, Spark, and Storm for data processing	
Adjunct Professor	New York University (<i>concurrent</i>)	New York, NY 2012 – 2013
	Taught the "Machine Learning" class.	
Adjunct Professor	Baruch College (<i>concurrent</i>)	New York, NY 2012 – 2013
	Taught the "Algorithm Design" class.	
Research Intern	Microsoft Research	Redmond, WA 2011 – 2011
	Projects on anomaly detection and time series modeling	

Honors and Awards

- Best Paper Award, Conference on Learning Theory (COLT), 2012
- Award of Excellence – Stars of Tomorrow, Microsoft Research, Asia
- Bachelor's Degree with Honors, Zhejiang University

Representative Publications

Full publication list: [Google Scholar](#)

Large Language Models (LLMs)

- APIGen-MT: Agentic Pipeline for Multi-Turn Data Generation via Simulated Agent-Human Interplay. [NeurIPS Datasets & Benchmarks Track](#), 2025. (*co-corresponding author*). [\[Data\]](#), [\[Model\]](#)
- APIGen: Automated Pipeline for Generating Verifiable and Diverse Function-Calling Datasets. [NeurIPS](#), 2024. [\[Data\]](#), [\[Model\]](#)
- xLAM: A Family of Large Action Models to Empower AI Agent Systems. [NAACL](#), 2024. (*co-corresponding author*). [\[Code\]](#)
- Retroformer: Retrospective Large Language Agents with Policy Gradient Optimization. [ICLR](#), 2024. (*co-corresponding author*).
- CodeGen: An Open Large Language Model for Code with Multi-Turn Program Synthesis. [ICLR](#), 2023. [\[Code\]](#)

AI Agents and Multi-Agent Systems

- BOLAA: Benchmarking and Orchestrating LLM-Augmented Autonomous Agents. [arXiv](#), 2023. [\[Code\]](#)
- AgentLite: A Lightweight Library for Building and Advancing Task-Oriented LLM Agent System. [arXiv](#), 2024. [\[Code\]](#)
- CRMArena: Understanding the Capacity of LLM Agents to Perform Professional CRM Tasks in Realistic Environments. [NAACL](#), 2025. [\[Code\]](#)
- ActionStudio: A Lightweight Framework for Data and Training of Large Action Models. [EMNLP](#), 2025.
- UserRL: Training Interactive User-Centric Agent via Reinforcement Learning. [arXiv](#), 2025. [\[Code\]](#)
- Enterprise Deep Research: Steerable Multi-Agent Deep Research for Enterprise Analytics. [arXiv](#), 2025.
- MCPEval: Automatic MCP-based Deep Evaluation for AI Agent Models. [arXiv](#), 2025. [\[Code\]](#)
- REX: Rapid Exploration and eXploitation for AI Agents. [arXiv](#), 2023.

LLM Reasoning, Routing, and Software Engineering

- Language Models are Hidden Reasoners: Unlocking Latent Reasoning Capabilities via Self-Rewarding. [arXiv](#), 2024.
- PRACT: Optimizing Principled Reasoning and Acting of LLM Agent. [CoNLL](#), 2024.
- LATTE: Learning to Think with Vision Specialists. [arXiv](#), 2024.
- ToolLibGen: Scalable Automatic Tool Creation and Aggregation for LLM Reasoning. [arXiv](#), 2025.
- xRouter: Training Cost-Aware LLMs Orchestration System via Reinforcement Learning. [arXiv](#), 2025.
- LoCoBench: A Benchmark for Long-Context Large Language Models in Complex Software Engineering. [arXiv](#), 2025. [\[Code\]](#)

Theory of Deep Learning and In-Context Learning

- Transformers as Statisticians: Provable In-Context Learning with In-Context Algorithm Selection. [NeurIPS](#), 2023.
- How do Transformers Learn In-Context Beyond Simple Functions? A Case Study on Learning with Representations. [ICLR](#), 2024.
- How Important is the Train-Validation Split in Meta-Learning? [ICML](#), 2021.
- Catastrophic Fisher Explosion: Early Phase Fisher Matrix Impacts Generalization. [ICML](#), 2021.

- [Ensemble of Averages: Improving Model Selection and Boosting Performance in Domain Generalization](#). [🔗](#) NeurIPS, 2022.

Reinforcement Learning

- [Policy Finetuning: Bridging Sample-Efficient Offline and Online Reinforcement Learning](#). [🔗](#) NeurIPS, 2021.
- [Sample-Efficient Learning of Stackelberg Equilibria in General-Sum Games](#). [🔗](#) NeurIPS, 2021.
- [WarpDrive: Extremely Fast End-to-End Deep Multi-Agent Reinforcement Learning on a GPU](#). [🔗](#) arXiv, 2021.
- [On the Generalization Gap in Reparameterizable Reinforcement Learning](#). [🔗](#) ICML, 2019.

Uncertainty Estimation & Reliability

- [Improved Online Conformal Prediction via Strongly Adaptive Online Learning](#). [🔗](#) ICML, 2023.
- [Understanding the Under-Coverage Bias in Uncertainty Estimation](#). [🔗](#) NeurIPS, 2021.
- [Don't Just Blame Over-parametrization for Over-confidence: Theoretical Analysis of Calibration in Binary Classification](#). [🔗](#) ICML, 2021.

Time Series, Causality and Neural Editing

- [Merlion: End-to-End Machine Learning for Time Series](#). [🔗](#) JMLR, 2023.
- [Causal Layering via Conditional Entropy](#). [🔗](#) PMLR, 2024.
- [Salesforce CausalAI Library: A Fast and Scalable Framework for Causal Analysis of Time Series and Tabular Data](#). [🔗](#) arXiv, 2023.
- [On the Unlikelihood of D-Separation](#). [🔗](#) arXiv, 2023.
- [Editing Arbitrary Propositions in LLMs without Subject Label](#). [🔗](#) arXiv, 2024.

Earlier Work in Machine Learning and Sparse Representation

- [Exact Recovery of Sparsely-Used Dictionaries](#). [🔗](#) COLT, 2012. **Best Paper Award**.
- [Adaptive Dropout with Rademacher Complexity Regularization](#). [🔗](#) ICLR, 2018.
- [Trace ratio vs. ratio trace for dimensionality reduction](#). [🔗](#) CVPR, 2007.

Open Source Projects

Large Action Models and AI Agents

- [xLAM & ActionStudio](#) [🔗](#): A family of large action models for AI agent systems, plus a lightweight framework for action-model data and training.
- [AgentLite](#) [🔗](#): A lightweight library for building task-oriented LLM agent systems.
- [BOLAA](#) [🔗](#): Benchmarking and orchestrating LLM-augmented autonomous agents.
- [CRMArena](#) [🔗](#): A benchmark for evaluating LLM agents on professional CRM tasks.
- [MCPEval](#) [🔗](#): Automatic MCP-based deep evaluation for AI agent models.
- [Enterprise Deep Research](#) [🔗](#): Steerable multi-agent deep research for enterprise analytics.
- [PersonaBench](#) [🔗](#): A benchmark for persona-based conversation systems.
- [MobileAIBench](#) [🔗](#): A benchmark for evaluating AI systems on mobile devices.

Function Calling and API Generation

- [APIGen](#) [🔗](#): Automated pipeline for generating verifiable and diverse function-calling datasets.
- [APIGen-MT](#) [🔗](#): Agentic pipeline for multi-turn data generation via simulated agent-human interplay.

Code Generation and Software Engineering

- [CodeGen](#) [🔗](#): A family of open-source models for code generation.
- [CoDA](#) [🔗](#): Coding language models via diffusion adaptation.
- [LoCoBench](#) [🔗](#): A benchmark for long-context large language models in complex software engineering.
- [Diversity Empowers Intelligence](#) [🔗](#): Integrating expertise of software engineering agents.

Vision and Multimodal Models

- [LATTE](#) [🔗](#): Learning to think with vision specialists.

- [xGen-MM \(BLIP3\)](#) : Multimodal generative models. [\[Code\]](#) 
- [UniControl](#) : A unified diffusion model for controllable image generation.
- [Hive](#) : Harnessing Human Feedback for Instructional Visual Editing.

Reinforcement Learning and Optimization

- [WarpDrive](#) : Extremely fast end-to-end deep multi-agent reinforcement learning on GPU.
- [UserRL](#) : User-centric reinforcement learning for interactive LLM agents.
- [UserBench](#) : An interactive gym environment for user-centric agents.
- [xRouter](#) : Training cost-aware LLM orchestration systems via reinforcement learning.

Prompt Optimization

- [Promptomatix](#) : A powerful framework for LLM prompt optimization.
- [Retroformer](#) : Retrospective large language agents with policy gradient optimization.

Time Series and Causal Learning

- [Merlion](#) : An easy-to-use library for time series anomaly detection and forecasting. [\[Blog\]](#) 
- [CausalAI](#) : A library for causal inference and causal discovery.

Dialog and Conversation Systems

- [Converse](#) : A framework for conversational AI applications.
- [DialogStudio](#) : A comprehensive dialogue understanding benchmark and toolkit.

Professional Service

- Reviewer: NeurIPS, ICML, ICLR, ACL, EMNLP, NAACL, AAAI, JMLR, TPAMI.

Skills

Technical: Machine Learning, Deep Learning, Reinforcement Learning, NLP, Computer Vision, Algorithms, AI Agents, Data Mining

Programming: Python

Frameworks & Tools: PyTorch, TensorFlow, Spark, Hadoop

Languages: English, Chinese